Introduction to Operations Research, 1

Lecture 14: Lagrangian Duality and Algorithms for Constrained NLP

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Weak and Strong Duality Theorem

Weak Duality Theorem

Primal Problem

$$p^* \triangleq \text{infimum}\{f(\mathbf{x}) : \mathbf{g}(\mathbf{x}) \leq \mathbf{0}, \mathbf{h}(\mathbf{x}) = \mathbf{0}, \mathbf{x} \in X\}$$

Lagrangian Dual Problem

$$d^* \triangleq \text{supremum}\{\theta(\mathbf{u}, \mathbf{v}) : \mathbf{u} \geq \mathbf{0}\}$$

where $\theta(\mathbf{u}, \mathbf{v}) = \inf\left\{f(\mathbf{x}) + \mathbf{u}^\top \mathbf{g}(\mathbf{x}) + \mathbf{v}^\top \mathbf{h}(\mathbf{x}) : \mathbf{x} \in X\right\}$.

Weak Duality Theorem

Let $\mathbf{x}$ be a feasible solution to the primal problem. Let $(\mathbf{u}, \mathbf{v})$ be a feasible solution to the dual problem. Then $f(\mathbf{x}) \geq \theta(\mathbf{u}, \mathbf{v})$.

Corollary

1. $p^* \geq d^*$;
2. *If $d^* = \infty$, then the primal problem is infeasible (dual unbounded $\rightarrow$ primal infeasible);*
3. *If $f(\bar{\mathbf{x}}) = \theta(\bar{\mathbf{u}}, \bar{\mathbf{v}})$, where $\bar{\mathbf{x}} \in S$ and $\bar{\mathbf{u}} \geq 0$, then $\bar{\mathbf{x}}$ and $(\bar{\mathbf{u}}, \bar{\mathbf{v}})$ respectively solves the primal and dual problems.*

If the strict inequality $p^* > d^*$ holds, we say there exists a duality gap.

Strong Duality Theorem

There is no duality gap for a convex program with Slater CQ.

Theorem

Let X be a convex set, $f : \mathbb{R}^n \to \mathbb{R}$ and $\mathbf{g} : \mathbb{R}^n \to \mathbb{R}^m$ be convex functions, and let $\mathbf{h} : \mathbb{R}^n \to \mathbb{R}^\ell$ be affine. In addition, suppose that the Slater constraint qualification holds, i.e., there exists an $\hat{\mathbf{x}} \in S$ such that $\mathbf{g}(\hat{\mathbf{x}}) < 0$, $\mathbf{h}(\hat{\mathbf{x}}) = 0$, and $\hat{\mathbf{x}} \in \text{int}X$. Then $p^* = d^*$, i.e., $\inf\{f(\mathbf{x}) : \mathbf{x} \in X, \mathbf{g}(\mathbf{x}) \leq 0, \mathbf{h}(\mathbf{x}) = 0\} = \sup\{\theta(\mathbf{u}, \mathbf{v}) : \mathbf{u} \geq 0\}$;

Corollary: complementary slackness condition

Under the above conditions, for the optimal solutions $\bar{\mathbf{x}}$ and $(\bar{\mathbf{u}}, \bar{\mathbf{v}})$ to the primal and dual problems respectively, we have $\bar{\mathbf{u}}^\top \mathbf{g}(\bar{\mathbf{x}}) = 0$.

Proof. Since $d^* = p^*$, we have $f(\bar{\mathbf{x}}) = \theta(\bar{\mathbf{u}}, \bar{\mathbf{v}}) = \inf\left\{f(\mathbf{x}) + \bar{\mathbf{u}}^\top \mathbf{g}(\mathbf{x}) + \bar{\mathbf{v}}^\top \mathbf{h}(\mathbf{x}) : \mathbf{x} \in X\right\}$. Since $g(\bar{\mathbf{x}}) \leq 0$ and $h(\bar{\mathbf{x}}) = 0$, we must have $f(\bar{\mathbf{x}}) \leq f(\bar{\mathbf{x}}) + \bar{\mathbf{u}}^\top \mathbf{g}(\bar{\mathbf{x}})$, which proves the result.

Usages of Lagrangian dual problem

1. **Identify near-optimal solutions**
   Since $f(\mathbf{x}) \geq \theta(\mathbf{u}, \mathbf{v})$ and weak duality theorem, a good dual solution can be used to bound the optimal values of the primal problem, and thus can be used to identify whether a primal solution is optimal or near optimal
2. **Identify optimal solutions**
   If further strong duality theorem holds under Slater CQ, one can prove the optimality of a primal solution by constructing a dual solution with the same objective value.
3. **Sensitivity analysis of primal solutions**
   The negative of the optimal Lagrangian multiplier values represents the rate of change in the optimal value with respect to the perturbations in the right-hand-side of the constraints (perturbation function)
4. **Algorithm development and convergence analysis**
   The dual problem has some useful mathematical and computational structure that can be exploited in the algorithmic development and convergence analysis.

Perturbation function

$$\begin{array}{ll} \text{minimize} & f(\mathbf{x}) \\ \text{subject to} & g_i(\mathbf{x}) \leq 0, \quad \text{for } i = 1, \ldots, m \\ & h_j(\mathbf{x}) = 0, \quad \text{for } j = 1, \ldots, \ell \\ & \mathbf{x} \in X \subseteq \mathbb{R}^n \end{array}$$

Definition

Define the perturbation function $p(\bullet) : \mathbb{R}^{m+\ell} \to \mathbb{R}$ as the optimal value function as follows with $\mathbf{y} = (y_1, \ldots, y_m, \ldots, y_{m+1}, \ldots, y_{m+\ell})$, i.e.,

$$p(\mathbf{y}) = \min \left\{ f(\mathbf{x}) : \begin{array}{l} g_i(\mathbf{x}) \leq y_i, \text{ for } i = 1, \ldots, m, \\ h_j(\mathbf{x}) = y_{j+m}, \text{ for } j = 1, \ldots, \ell, \\ \mathbf{x} \in X \end{array} \right\}$$

Perturbation function

Convexity

Suppose that f and $\{g_i\}$ are convex functions, and $\{h_j\}$ is affine.

- $p(\bullet)$ is a convex function.
- In addition, suppose that $\bar{\mathbf{x}}$ and $(\bar{\mathbf{u}}, \bar{\mathbf{v}})$ are optimal solutions to the primal and dual problems with $p^* = d^*$. Then $-(\bar{\mathbf{u}}, \bar{\mathbf{v}}) \in \partial p(\mathbf{0})$ and $p(\mathbf{y}) \geq p(\mathbf{0}) - (\bar{\mathbf{u}}^\top, \bar{\mathbf{v}}^\top)\mathbf{y}, \forall \mathbf{y} \in \mathbb{R}^{m+\ell}$.

Corollary: differentiable case

Under the conditions above, if p is continuously differentiable at $\mathbf{y} = 0$, then

$$\nabla p(\mathbf{0}) = -\begin{pmatrix} \bar{\mathbf{u}} \\ \bar{\mathbf{v}} \end{pmatrix}.$$

Interpretations:

- the negative of the optimal Lagrangian multiplier give the marginal rates of change in the optimal objective value of the primal problem with respect to perturbations in the RHS;
- economic interpretations in terms of the marginal change in the optimal costs with respect to perturbations in resources such as material, labor, and demand in the constraints.

Dual function

Dual function

Dual function $\theta(\mathbf{u}, \mathbf{v}) = \min\{f(\mathbf{x}) + \mathbf{u}^\top \mathbf{g}(\mathbf{x}) + \mathbf{v}^\top \mathbf{h}(\mathbf{x}) : \mathbf{x} \in X\}$.

Example:

$$\begin{array}{ll} \text{minimize} & -2x_1 + x_2 \\ \text{subject to} & x_1 + x_2 - 3 = 0 \\ & (x_1, x_2) \in X \triangleq \{(0,0), (0,4), (4,4), (4,0), (1,2), (2,1)\} \end{array}$$

primal solution $(x_1^*, x_2^*) = (2, 1)$ and $p^* = -3$

dual function

$$\begin{aligned} \theta(v) &= \min\{(-2x_1 + x_2) + v(x_1 + x_2 - 3) : (x_1, x_2) \in X\} \\ &= \min\{-3v, 4 + v, -4 + 5v, -8 + v, 0, -3\} \end{aligned}$$

Solving the dual problem

- The dual problem is to maximize a concave dual function with nonnegative constraints on partial Lagrangian multiplier $\mathbf{u}$;
- The dual problem may have smaller dimensions and simpler constraints;
- If there is no duality gap and the dual is solved exactly with the multiplier $(\mathbf{u}^*, \mathbf{v}^*)$, the optimal primal solution can be obtained by minimizing the Lagrangian function $L(\mathbf{x}; \mathbf{u}^*, \mathbf{v}^*)$ over $\mathbf{x} \in X$;
- To solve the dual problem $\boxed{\text{maximize}_{\mathbf{u} \geq 0}\ \theta(\mathbf{u}, \mathbf{v})}$, we need to understand the dual function θ

Dual function

Suppose X is a nonempty compact set in $\mathbb{R}^n$.

Properties of the dual function $\theta(\mathbf{u}, \mathbf{v})$:

- θ is concave (Excercise);
- under what condition θ is differentiable, and what is the gradient?
- how to characterize its subdifferential $\partial\theta(\bar{\mathbf{u}}, \bar{\mathbf{v}}) \triangleq -\partial(-\theta(\bar{\mathbf{u}}, \bar{\mathbf{v}}))$

Dual function: subdifferential and differentiability

Lagrangian Dual Problem

$$d^* \triangleq \begin{array}{ll} \text{maximize} & \theta(\mathbf{u}, \mathbf{v}) \\ \text{subject to} & \mathbf{u} \geq \mathbf{0} \end{array}$$

where $\theta(\mathbf{u}, \mathbf{v}) = \inf \left\{ f(\mathbf{x}) + \mathbf{u}^\top \mathbf{g}(\mathbf{x}) + \mathbf{v}^\top \mathbf{h}(\mathbf{x}) : \mathbf{x} \in X \right\}$

Theorem

Suppose X is a nonempty compact set in $\mathbb{R}^n$, $f(\bullet)$, $g(\bullet)$ and $h(\bullet)$ are continuous functions. For the function $\theta(\mathbf{u}, \mathbf{v})$,

1. $\theta(\bullet, \bullet)$ is a concave function;

2. given $(\bar{\mathbf{u}}, \bar{\mathbf{v}}) \in \mathbb{R}^{m+\ell}$, if $\bar{\mathbf{x}} \in \text{argmin}\{f(\mathbf{x}) + \bar{\mathbf{u}}^\top \mathbf{g}(\mathbf{x}) + \bar{\mathbf{v}}^\top \mathbf{h}(\mathbf{x}) : \mathbf{x} \in X\}$, then $(\mathbf{g}(\bar{\mathbf{x}}), \mathbf{h}(\bar{\mathbf{x}})) \in \partial\, \theta(\bar{\mathbf{u}}, \bar{\mathbf{v}})$;

3. given $(\bar{\mathbf{u}}, \bar{\mathbf{v}}) \in \mathbb{R}^{m+\ell}$, if $\{\bar{\mathbf{x}}\} = \text{argmin}\{f(\mathbf{x}) + \bar{\mathbf{u}}^\top \mathbf{g}(\mathbf{x}) + \bar{\mathbf{v}}^\top \mathbf{h}(\mathbf{x}) : \mathbf{x} \in X\}$, then θ is differentiable at $(\bar{\mathbf{u}}, \bar{\mathbf{v}})$, and $\nabla\, \theta(\bar{\mathbf{u}}, \bar{\mathbf{v}}) = (\mathbf{g}(\bar{\mathbf{x}}), \mathbf{h}(\bar{\mathbf{x}}))$.

Solving the dual problem

Lagrangian Dual Problem

$$d^* \triangleq \quad \text{maximize} \quad \theta(\mathbf{u}, \mathbf{v})$$
$$\text{subject to} \quad \mathbf{u} \geq \mathbf{0}$$

where $\theta(\mathbf{u}, \mathbf{v}) = \inf \left\{ f(\mathbf{x}) + \mathbf{u}^\top \mathbf{g}(\mathbf{x}) + \mathbf{v}^\top \mathbf{h}(\mathbf{x}) : \mathbf{x} \in X \right\}$

How to solve the dual problem?

A variant of projected (sub)gradient method:

- if the new iterate point satisfies the nonnegative constraint, then we are fine
- otherwise, project the iterate point back to the nonnegative constraint set; namely, set the negative element of the solution vector as zero.

Projected subgradient/gradient method with $(\mathbf{u}^r, \mathbf{v}^r)$.

$$\mathbf{x}^r \in \text{argmin} \left\{ f(\mathbf{x}) + (\mathbf{u}^r)^\top \mathbf{g}(\mathbf{x}) + (\mathbf{v}^\mathbf{r})^\top \mathbf{h}(\mathbf{x}) : \mathbf{x} \in X \right\},$$
$$\mathbf{u}^{r+1} = \left[\mathbf{u}^r + \alpha^r \mathbf{g}(\mathbf{x}^r) \right]_+, \quad \mathbf{v}^{r+1} = \mathbf{v}^r + \alpha^r \, \mathbf{h}(\mathbf{x}^r)$$

Solving the dual problem

The **dual problem** over $(\mathbf{u}, \mathbf{v})$

$$\begin{array}{ll} \text{maximize} & \theta(\mathbf{u}, \mathbf{v}) = \inf\{f(\mathbf{x}) + \mathbf{u}^\top \mathbf{g}(\mathbf{x}) + \mathbf{v}^\top \mathbf{h}(\mathbf{x}) : \mathbf{x} \in X\} \\ \text{subject to} & \mathbf{u} \geq \mathbf{0} \end{array}$$

$\iff$ a maximization problem over $(\mathbf{u}, \mathbf{v}, z)$ with infinitely many constraints.

$$\begin{array}{ll} \text{maximize} & z \\ \text{subject to} & z \leq f(\mathbf{x}) + \mathbf{u}^\top \mathbf{g}(\mathbf{x}) + \mathbf{v}^\top \mathbf{h}(\mathbf{x}), \quad \forall \mathbf{x} \in X \\ & \mathbf{u} \geq \mathbf{0} \end{array}$$

The **approximating problem** given points $\mathbf{x}^1, \ldots, \mathbf{x}^{k-1}$,

$$\begin{array}{ll} \text{maximize} & z \\ \text{subject to} & z \leq f(\mathbf{x}^j) + \mathbf{u}^\top \mathbf{g}(\mathbf{x}^j) + \mathbf{v}^\top \mathbf{h}(\mathbf{x}^\mathbf{j}), \quad \forall j = 1, \ldots, k-1 \\ & \mathbf{u} \geq \mathbf{0} \end{array}$$

$$\iff \begin{array}{ll} \text{maximize} & \hat{\theta}^{k-1}(\mathbf{u}, \mathbf{v}) \triangleq \min\{f(\mathbf{x}^j) + \mathbf{u}^\top \mathbf{g}(\mathbf{x}^j) + \mathbf{v}^\top \mathbf{h}(\mathbf{x}^\mathbf{j}) : j \in [k-1]\} \\ \text{subject to} & \mathbf{u} \geq \mathbf{0} \end{array}$$

Cutting plane method

Cutting plane method: iteratively construct a concave piecewise-linear function $\hat{\theta}^{k-1}(\bullet,\bullet)$ to approximate the dual function $\theta(\bullet,\bullet)$, which is refined by progressively adding linear cuts.

Start with a feasible solution $\mathbf{x}_0$.
For $k = 1, 2, \ldots,$

1. given $\{\mathbf{x}^j\}_{j=1}^{k-1}$, solve the following problem and obtain the solution $(z^k, \mathbf{u}^k, \mathbf{v}^k)$;

$$\begin{array}{ll} \text{maximize} & z \\ \text{subject to} & z \leq f(\mathbf{x}^j) + \mathbf{u}^\top \mathbf{g}(\mathbf{x}^j) + \mathbf{v}^\top \mathbf{h}(\mathbf{x}^\mathbf{j}), \quad \forall\, j = 1, \ldots, k-1 \\ & \mathbf{u} \geq \mathbf{0} \end{array}$$

2. compute $\theta(\mathbf{u}^k, \mathbf{v}^k)$, and $\mathbf{x}^k \in \operatorname{argmin} \left\{ f(\mathbf{x}) + (\mathbf{u}^k)^\top \mathbf{g}(\mathbf{x}) + (\mathbf{v}^\mathbf{k})^\top \mathbf{h}(\mathbf{x}) : \mathbf{x} \in X \right\}$,
3. if $\theta(\mathbf{u}^k, \mathbf{v}^k) = z^k$, then $(\mathbf{u}^k, \mathbf{v}^k)$ is an optimal solution to the dual problem; otherwise, continue to the next iteration.

Convergence: Assume that f, $\mathbf{g}$, $\mathbf{h}$ are continuous and that X is compact. Then any limit point of a sequence $\{(\mathbf{u}^k, \mathbf{v}^k)\}$ generated by the cutting plane method is a dual optimal solution.

Algorithms for constrained nonlinear programs

Projected gradient method

$$\boxed{\underset{\mathbf{x}\in X\subseteq\mathbb{R}^n}{\text{minimize}} \quad f(\mathbf{x})}$$

$$\mathbf{x}^{r+1} = \mathcal{P}_X\left(\mathbf{x}^r - t\nabla f(\mathbf{x}^r)\right)$$

Observations:

• the iterative process maintains the feasibility of the iterates;

• the constrained set X needs to be simple enough for the projection operation, such as a box constraint ($l_i \leq \mathbf{x}_i \leq u_i$ for $i = 1, \ldots, n$), a Euclidean ball constraint ($\|\mathbf{x}\|_2 \leq r$), a hyperplane constraint ($\mathbf{a}^\top\mathbf{x} = b$);

• If X is not simple enough to obtain the projection step, then one may consider an extra inner-loop algorithm to solve the projection problem

Projected subgradient method

$$\boxed{\underset{\mathbf{x}\in X\subseteq\mathbb{R}^n}{\text{minimize}} \quad f(\mathbf{x})}$$

Suppose $f(\bullet)$ is convex, and X is closed and convex.

Projected subgradient method

$$\mathbf{x}^{r+1} = \mathcal{P}_X\left(\mathbf{x}^r - \alpha^r \mathbf{g}^r\right), \quad \text{with } \mathbf{g}^r \in \partial f(\mathbf{x}^r)$$

the solution sequence provided by the projected subgradient method converges in terms of global optimality with diminishing step size $\{\alpha^r\}$ satisfying $\lim_{r\to\infty} \alpha^r = 0$ and $\sum_{r=0}^{\infty} \alpha^r + \infty$.

Frank-Wolfe method

Consider minimize $\{ f(\mathbf{x}) : \mathbf{x} \in X \subseteq \mathbb{R}^n \}$. Suppose

- $f(\bullet)$ is a convex differentiable function with Lipschitz gradient;
- X is a compact convex set, and easy to compute with.

Frank-Wolfe method:

$$\boxed{\mathbf{z}^{r+1} \in \operatorname*{argmin}_{\mathbf{x} \in X} f(\mathbf{x}^r) + \nabla f(\mathbf{x}^r)^\top (\mathbf{x} - \mathbf{x}^r); \quad \mathbf{x}^{r+1} = (1 - \alpha^r)\mathbf{x}^r + \alpha^r \mathbf{z}^{r+1}}$$

Observations:

- the feasibility of the iterates is maintained throughout the iteration process, by convexity for any choice $0 \leq \alpha^r \leq 1$;

- update step is obtained by minimizing a linear function over the constraint set X, which is often simpler or cheaper than the projection operation in the projected gradient method;

- the default choice for step sizes is $\alpha^r = 2/(r+1)$ for $r = 0, 1, 2, \ldots$; no dependence on Lipschitz constant;

Extension: location problems

Set covering problems

Set covering problem

Each activity possesses some but not all of the characteristics. The objective is to determine the least costly combination of activities that collectively possess (cover) each characteristic at least once.

Thus, let S_i be the set of all activities that possess characteristic i. At least one member of the set S_i must be included among the chosen activities, so a constraint is $\sum_{j \in S_i} x_j \geq 1$.

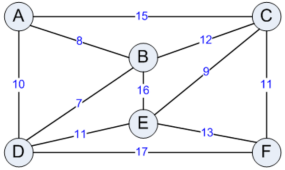
A set covering problem is of the form

$$\begin{array}{ll} \text{minimize} & \mathbf{c}^\top \mathbf{x} \\ \text{s.t.} & \sum_{j \in S_i} x_j \geq 1, \forall i = 1, \ldots, p \\ & x_j \in \{0, 1\}, \forall j = 1, \ldots, k \end{array}$$

Set covering problem

Problem description

- Graph: $G = (N, A)$
- Set of demand nodes to be covered: $I \subset N$
- Set of candidate facility sites: $J \subset N$
- Cost to locate a facility at candidate site j: f_j for $j \in J$
- Distance of arc $(i, j) \in A$: d_{ij}
- Coverage distance: C



Coverage distance: 11

$$a_{ij} = \begin{cases} 1 & \text{if candidate site } j \text{ covers demand node } i\text{, i.e., } d_{ij} \leq C \\ 0 & \text{otherwise} \end{cases}$$

Goal: Locate a cost-minimizing set of facilities that cover all the demand nodes

Set covering problem

Decision variables:

$$x_j = \begin{cases} 1 & \text{if we locate a facility at candidate site } j \\ 0 & \text{otherwise} \end{cases}$$

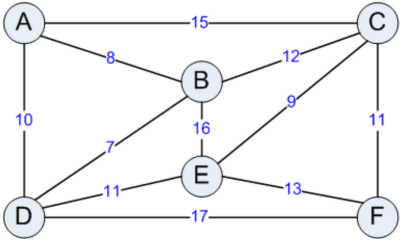
Model:

minimize	$\sum_{j \in J} f_j x_j$	minimize the total facility cost
subject to	$\sum_{j} a_{ij} x_j \geq 1, \forall i \in I$	each demand node is covered by at least one facility
	$x_j \in \{0, 1\}, \forall j \in J$	

Set covering problem

Model:

$$
\begin{array}{ll}
\text{minimize} & \displaystyle\sum_{j \in J} f_j x_j \\
\text{subject to} & \displaystyle\sum_{j} a_{ij} x_j \geq 1, \forall i \in I \\
& x_j \in \{0,1\}, \forall j \in J
\end{array}
$$

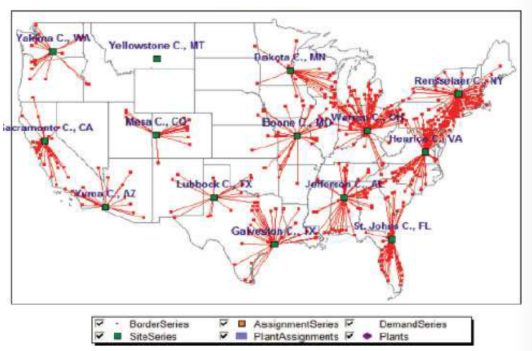


Coverage distance: 11

$$
\begin{array}{ll}
\text{minimize} & x_A + x_B + x_C + x_D + x_E + x_F \\
\text{subject to} & x_A + x_B + x_D \geq 1 \quad (A) \\
& x_A + x_B + x_D \geq 1 \quad (B) \\
& x_C + x_E + x_F \geq 1 \quad (C) \\
& \cdots \\
& x_A, x_B, x_C, x_D, x_E, x_F \in \{0,1\}
\end{array}
$$

Set covering problem: example

- 500 largest counties in the U.S., 75% of the U.S. population
- Coverage distance: 300 miles
- 14 facility sites achieves 100% coverage



[1]Source: Daskin (2008)

Maximum covering problem

Maximum covering problem

Problem description

- Graph: $G = (N, A)$
- Set of demand nodes to be covered: $I \subset N$
- Set of candidate facility sites: $J \subset N$
- Maximum number of facilities: P
- Demand at node i: h_i
- Cost to locate a facility at candidate site j: f_j for $j \in J$
- Distance of arc $(i, j) \in A$: d_{ij}
- Coverage distance: C

$$a_{ij} = \begin{cases} 1 & \text{if candidate site } j \text{ covers demand node } i\text{, i.e., } d_{ij} \leq C \\ 0 & \text{otherwise} \end{cases}$$

Goal: Locate a set of facilities that maximize the covered demands

Maximum covering problem

Decision variables:

$$x_j = \begin{cases} 1 & \text{if we locate a facility at candidate site } j \\ 0 & \text{otherwise} \end{cases}$$

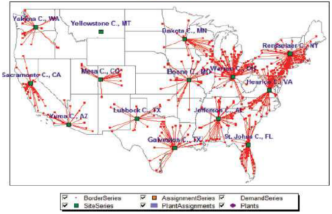
$$z_i = \begin{cases} 1 & \text{if the demand node } i \in I \text{ is covered by a facility} \\ 0 & \text{otherwise} \end{cases}$$

Model:

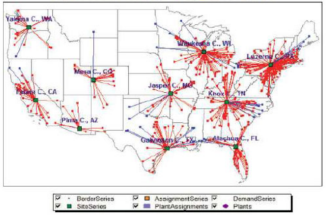
maximize	$\sum_{i \in I} h_i z_i$	maximize the covered demands
subject to	$\sum_j a_{ij} x_j \geq z_i, \forall i \in I$	each demand node is covered by at least one
	$\sum_j x_j \leq P,$	maximum number of facilities
	$x_j \in \{0,1\}, z_i \in \{0,1\}, \forall j \in J, \forall i \in I$	

Maximum covering problem

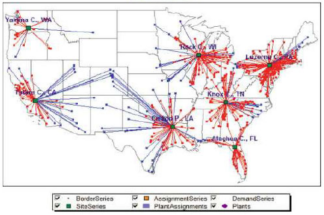
- 500 largest counties in the U.S., 75% of the U.S. population
- Coverage distance: 300 miles
- 14 facility sites achieves 100% coverage
- 10 facility sites achieves 97% coverage
- 7 facility sites achieves 88% coverage



14 sites: 100% coverage



10 sites: 97% coverage



7 sites: 88% coverage

[1]Source: Daskin (2008)

Set Covering Problem vs Maximum Covering problem

Set Covering Problem (100 % coverage) :

$$\begin{aligned} \text{minimize} \quad & \sum_{j \in J} f_j x_j \\ \text{subject to} \quad & \sum_{j \in J} a_{ij} x_j \geq 1, \forall i \in I \\ & x_j \in \{0,1\}, \forall j \in J \end{aligned}$$

Maximum Covering problem :

$$\begin{aligned} \text{maximize} \quad & \sum_{i \in I} h_i z_i \\ \text{subject to} \quad & \sum_{j \in J} a_{ij} x_j \geq z_i, \forall i \in I \\ & \sum_{j \in J} x_j \leq P, \\ & x_j \in \{0,1\}, z_i \in \{0,1\}, \forall j \in J, \forall i \in I \end{aligned}$$